Amazon Web Services: Big Data and Data Streaming

Topics

- Exploring Large Redshift Databases
- Processing Data with EMR (Elastic MapReduce)
- Streaming Data with Kinesis
- Amazon Athena and AWS Glue
- Visualizing Data with QuickSight
- Moving Transferred Data Using AWS Data Pipeline
- Implementing Amazon Managed Streaming for Apache Kafka
- Data Streaming vs. Batch Processing

Exploring Large Redshift Databases

The 3 V's of Big Data

Volume

Ranges from terabytes to petabytes of data

Variety

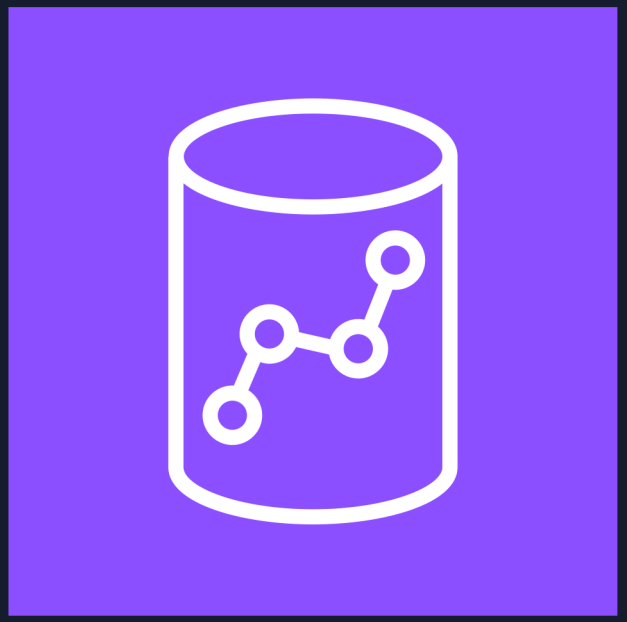
Includes data from a wide range of sources and formats

Velocity

Data must be collected, stored, processed, and analyzed within a short period of time

What is Redshift?

- Redshift is a fully managed, petabyte-scale data warehouse service in the cloud
- It’s a very large relational database traditionally used in big data applications



Amazon Redshift

Redshift Overview and Uses

Size

- Redshift is incredibly big - it can hold up to 16 PB of data. You don't have to split up your large datasets

Relational

- This database is relational. You use your standard SQL and business intelligence (BI) tools to interact with it

Based on PostgreSQL

- Based on PostgreSQL database engine type; however, it is not used for OLTP workloads

Usage

- Not a replacement for standard RDS databases

High Performance

- Offers up to 10x the performance of other data warehouses in the cloud

Columnar

- Storage of data is column-based instead of row-based. Allows for efficient parallel queries

High Availability, Snapshots, and Disaster Recovery

- Redshift now supports Multi-AZ deployments
  - It only spans two AZs currently
- Snapshots are incremental and point-in-time
  - They can be automated or manual
  - Always contained in Amazon S3
- No conversions from Single-AZ to Multi-AZ or vice versa

Redshift Spectrum and Enhanced VPC Routing

Redshift Spectrum

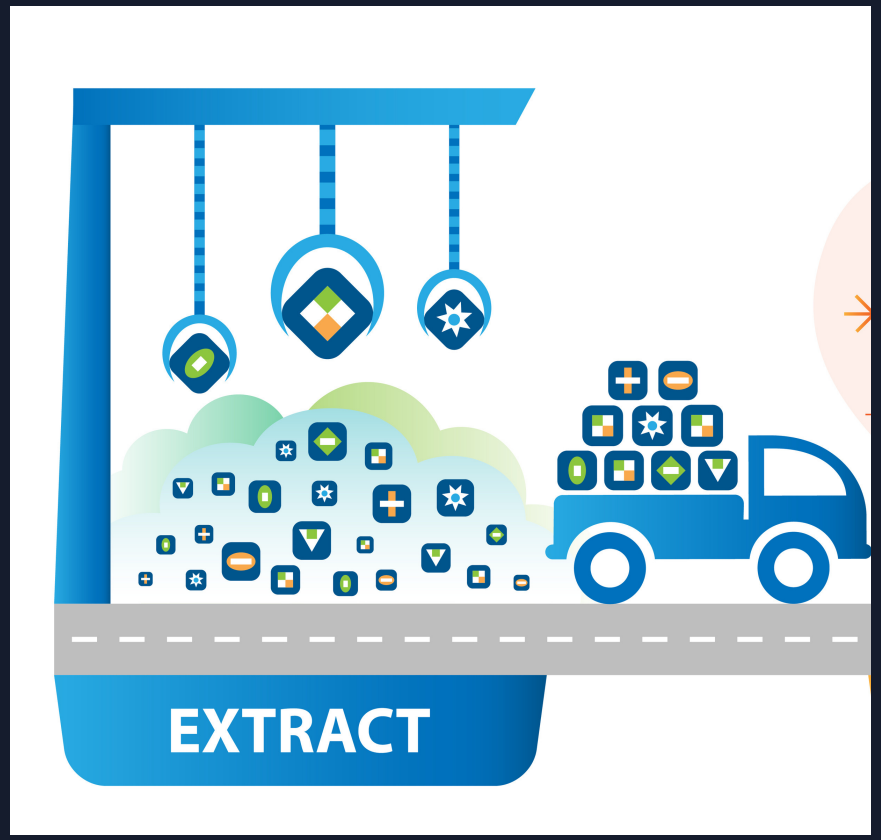
- Efficiently query and retrieve data from Amazon S3 without having to load data into Amazon Redshift table

Enhanced VPC Routing

- All COPY and UNLOAD traffic between your cluster and your data repositories is forced through your VPC

Processing Data with EMR (Elastic MapReduce)

What is ETL?



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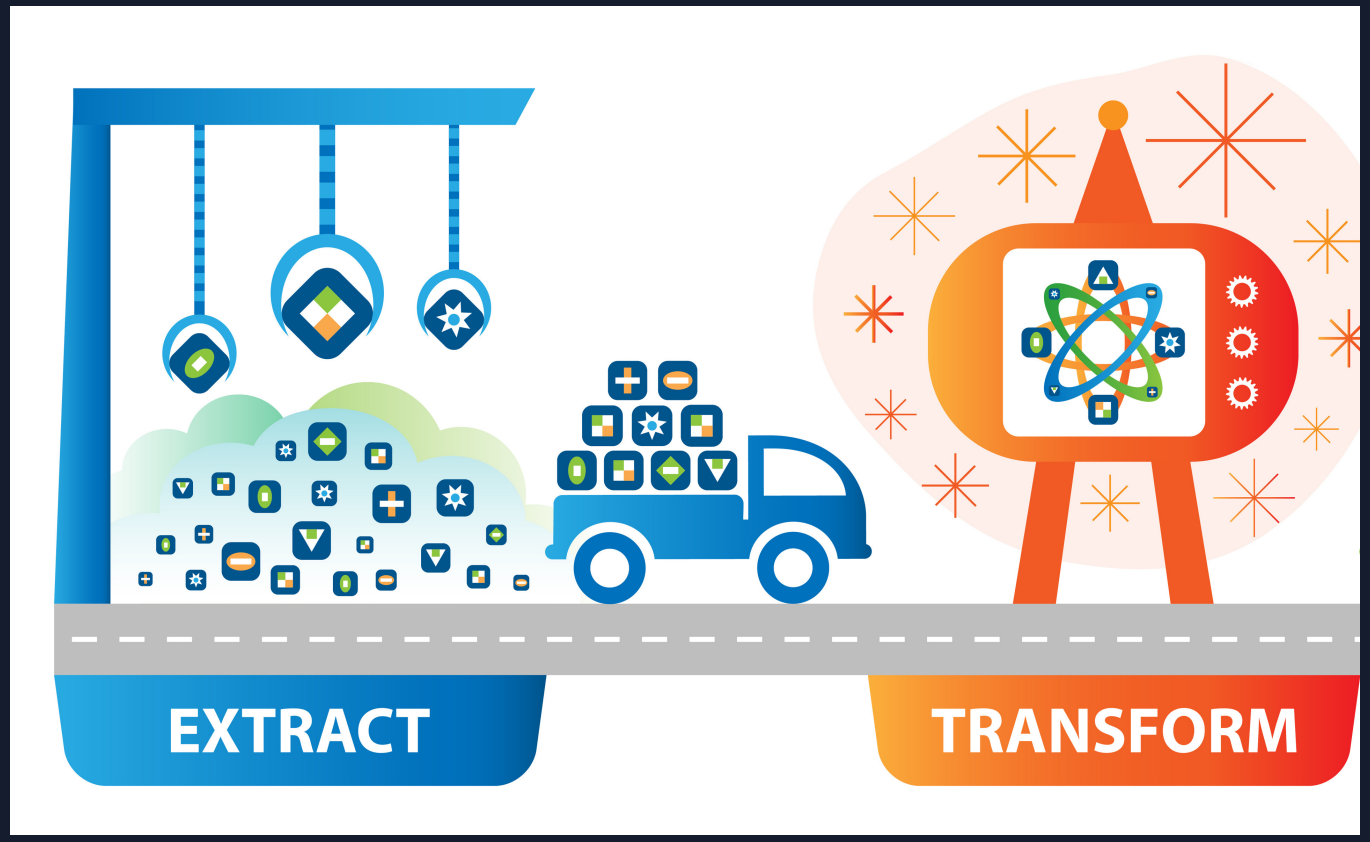
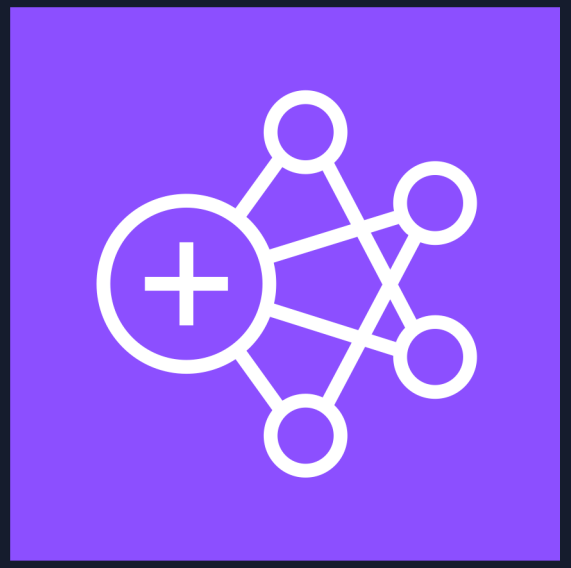


What is ETL?

What is Amazon Elastic MapReduce (EMR)?

- An AWS service to help with ETL processing
- A managed big data platform to process vast amounts of data using open-source tools, such as Spark, Hive, HBase, Flink, Hudi, and Presto



Amazon EMR

EMR Storage

Hadoop Distributed File System (HDFS)	EMR File System (EMRFS)	Local File System

Amazon EMR Clusters and Nodes

Primary Node

Manages the cluster, coordinates distribution of data and tasks, track health statuses

Core Node

Runs tasks and stores data in HDFS; long-running!

Task Node

ONLY runs tasks, with no storage of data within HDFS. Optional; typically, Spot instances

Purchasing Options and Cluster Types

1

On-Demand

Most reliable option. Will not be terminated. Most expensive choice.

3

Spot

Cheapest option. Terminated without warning. Used for **task nodes**.

2

Reserved

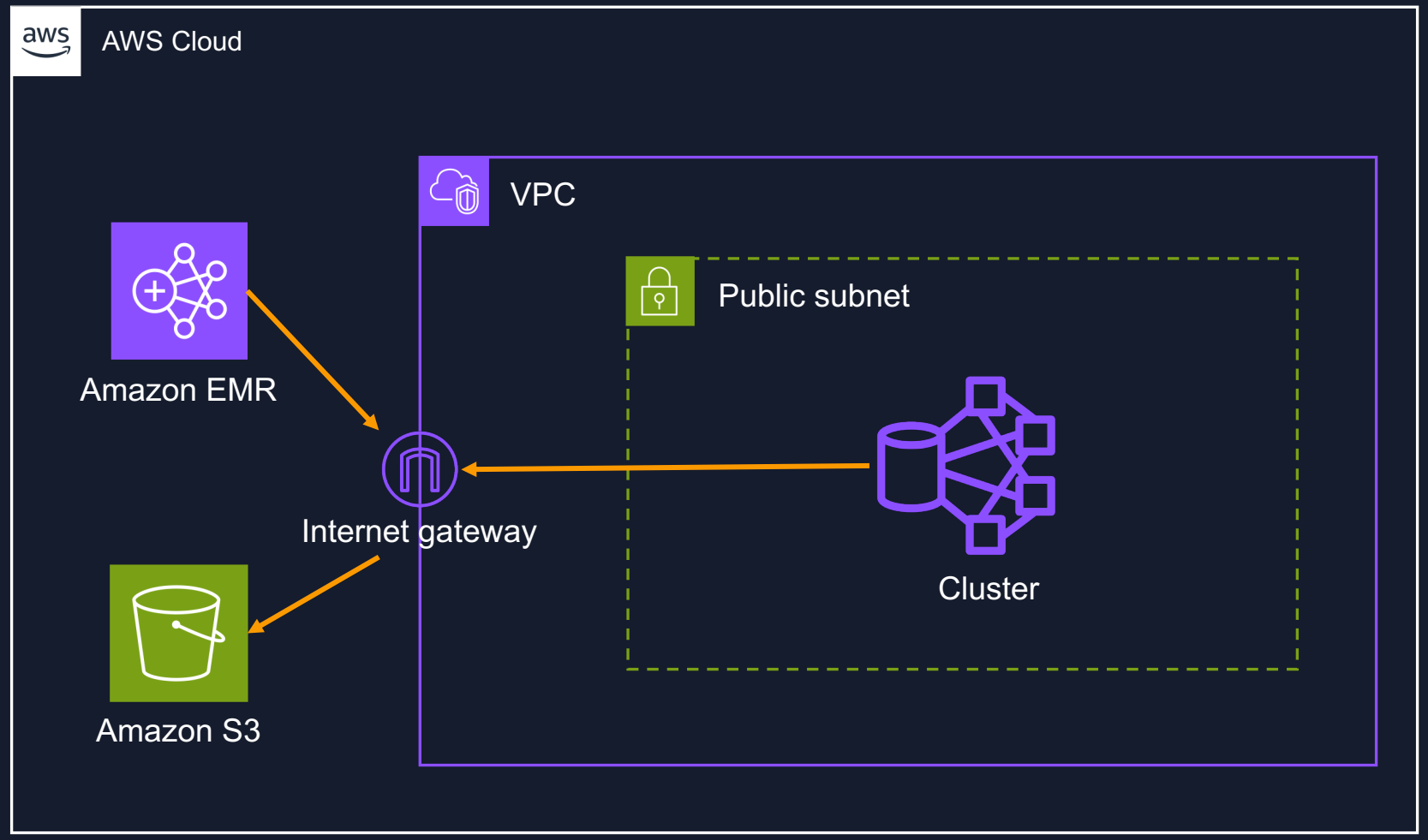
Minimum of 1 year. Great cost savings. Used for **primary nodes** and **core nodes**.

4

Long-running or Temporary?

Clusters can be either **long-running** or **transient** (temporary)

EMR Architecture



Streaming Data with Kinesis

What is Kinesis?

- Kinesis allows you to ingest, process, and analyze real-time streaming data

- Like huge data highway connected to your AWS account

Amazon Kinesis

2 Versions: Data Streams and Data Firehose

Data Streams

Purpose

- Real-time streaming for ingesting data

Speed

- Real time

Difficulty

- You're responsible for creating the consumer and scaling the stream.

Data Firehose

Purpose

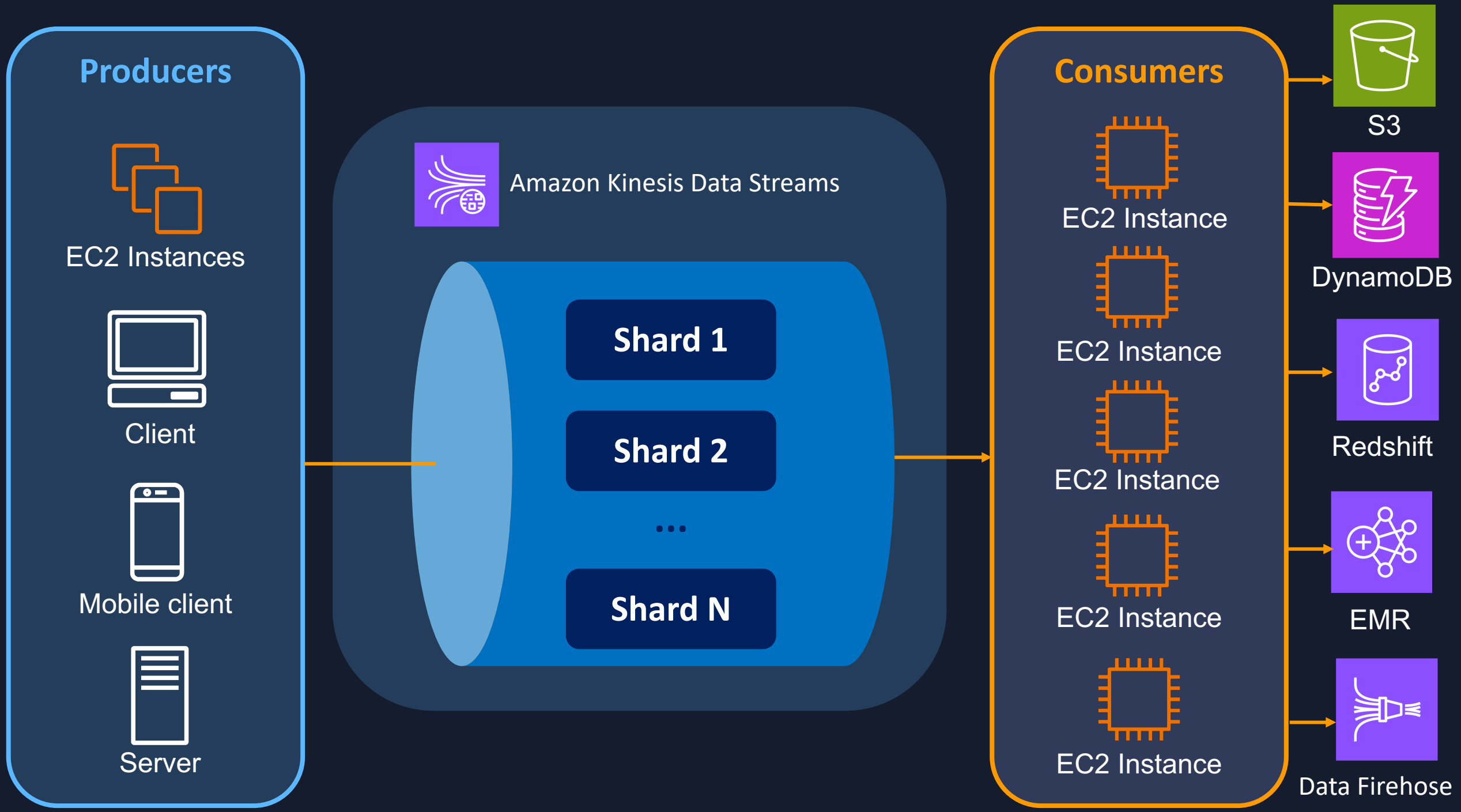
- Data transfer tool to get information to S3, Redshift, Elasticsearch, or Splunk

Speed

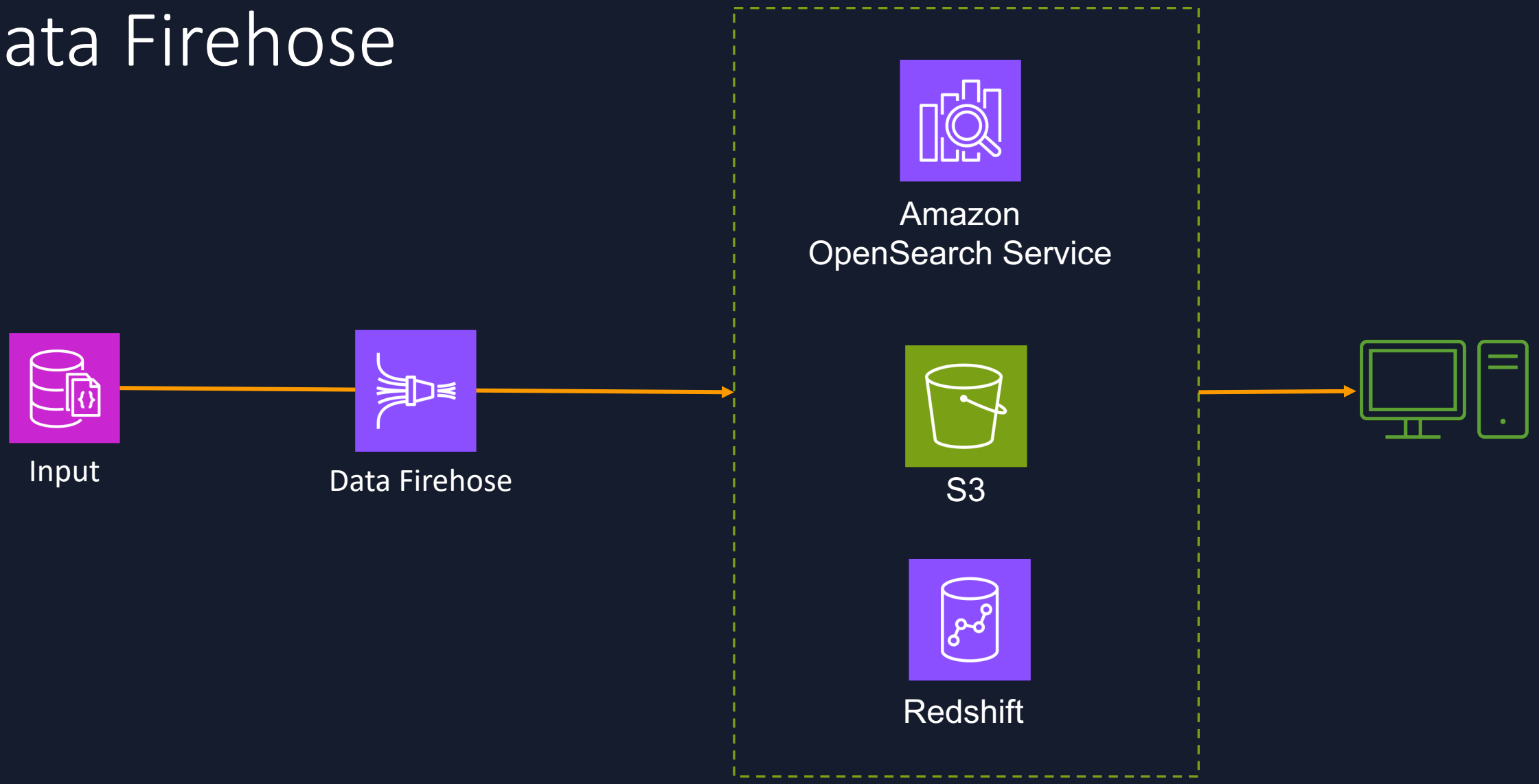
- Near real time (within 60 seconds)

Difficulty

- Plug and play with AWS architecture


Producers
EC2 Instances
Client
Mobile client
Server
Amazon Kinesis Data Streams
Shard 1
Shard 2
...
Shard N
Consumers
EC2 Instance
EC2 Instance
EC2 Instance
EC2 Instance
EC2 Instance
S3
DynamoDB
Redshift
EMR
Data Firehose

Data Firehose



Kinesis Data Analytics

Easy

- It's very simple to tie Data Analytics into your Kinesis pipeline
- It's directly supported by Data Firehose and Data Streams

No Servers

- Fully-managed real-time serverless service
- It will automatically handle scaling and provisioning of needed resources

Cost

- You only pay for the amount of resources you consume as your data passes through

SQS vs. Kinesis

SQS

- SQS is a messaging broker that is simple to use and doesn't require much configuration
- It doesn't offer real-time message delivery

VS

Kinesis

- Kinesis is a bit more complicated to configure and is mostly used in big data applications
- It does provide real-time communication

Amazon Athena and AWS Glue

What is Athena?

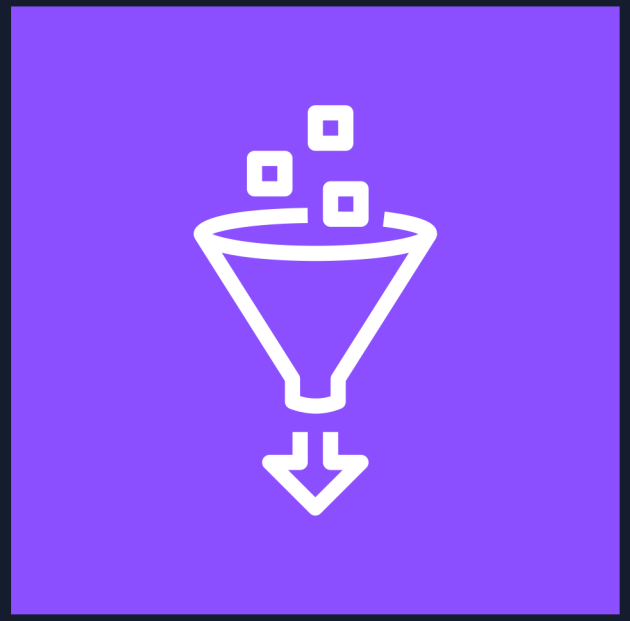
- Athena is an interactive query service that makes it easy to analyze data in S3 using SQL
- This allows you to directly query data in your S3 bucket without loading it into a database

Amazon Athena

What is Glue?

- Glue is a serverless data integration service that makes it easy to discover, prepare, and combine data

- It allows you to perform ETL workloads without managing underlying servers



AWS Glue

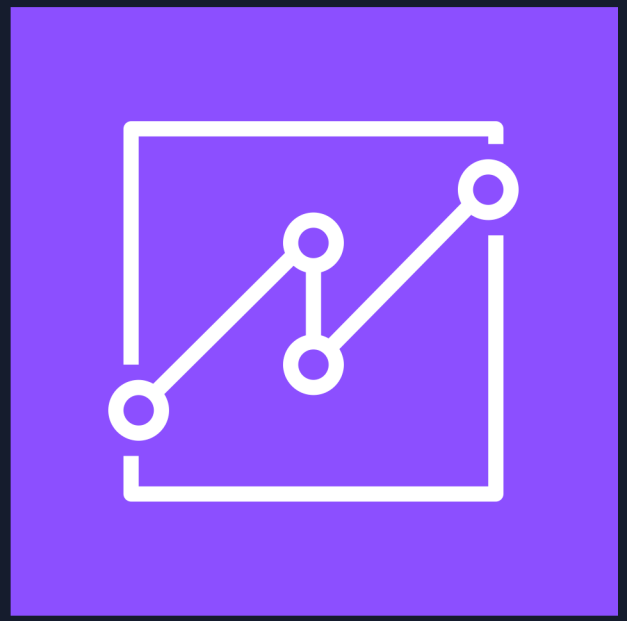
Architecture Diagram

Visualizing Data with QuickSight

What is QuickSight

- Amazon QuickSight is a fully managed business intelligence (BI) data visualization service

- It allows you to easily create dashboards and share them within your company



Amazon QuickSight

Features and Use Cases

- Useful for business data visualization, ad-hoc data analytics, and obtaining important data-based business insights
- Integrates with Amazon RDS, Amazon Aurora, Athena, S3m and many more
- SPICE: Robust in-memory engines used to perform advanced calculations
- Enterprise offering allows you to implement Column-Level Security (CLS)
- Pricing is on a per-session and per-user basis

Dashboards, Users, and Groups

Users can be created for accessing QuickSight

Enterprise version allows you to create groups

Users and groups only exist in QuickSight

Dashboards allow for stored configurations and filtering

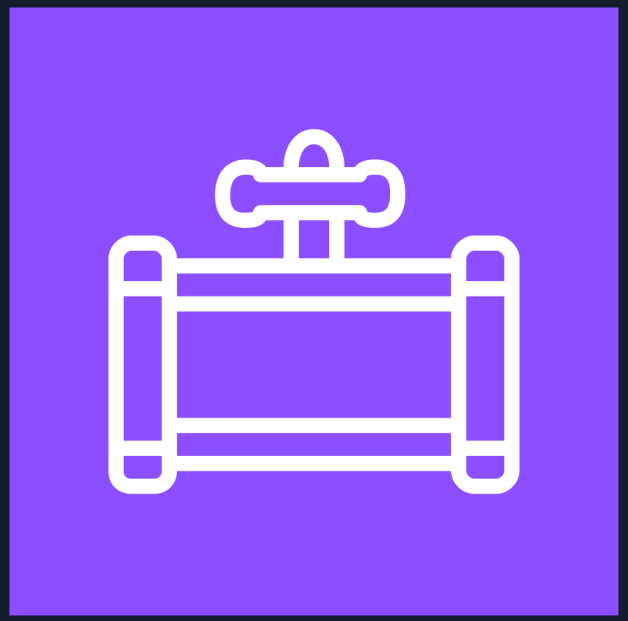
Can share dashboards and analysis results with users and groups

Architecture Diagram

Moving Transferred Data Using AWS Data Pipeline

What is AWS Data Pipeline

- AWS Data Pipeline is a managed Extract, Transform, Load (ETL) service for automating movement and transformation of your data



AWS Data Pipeline

AWS Data Pipeline Overview

Data Drive
Define data-driven workflows. Steps are dependent on previous tasks completing successfully

Parameters
Define your parameters for data transformations. Data Pipeline enforces your chosen logic

Highly Available
AWS hosts the infrastructure on highly available and distributed infrastructures

Handling Failures
Automatically retries failed activities. Configure notifications via SNS for failures or successful tasks

AWS Storage Service
Integrates with DynamoDB, RDS, Redshift, and S3

AWS Compute
Works with EC2 and EMR for compute needs

Components

1 **Pipeline Definition**
Specify business logic of data management needs

3 **Task Runners**
EC2 – poll for different tasks and perform them when found

2 **Managed Compute**
Service will create EC2 instances to perform activities – or use existing EC2

4 **Data Nodes**
Define locations and types of data that will be input and output

Popular Use Cases

- Processing data in EMR using Hadoop streaming
- Importing or exporting DynamoDB data
- Copying CSV files or data between S3 buckets
- Exporting RDS data to S3
- Copying data to Redshift

Implementing Amazon Managed Streaming for Apache Kafka

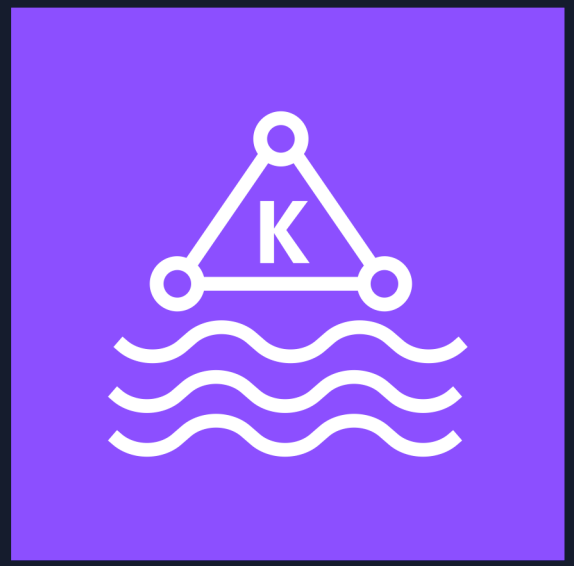
Amazon MSK Overview

Apache Kafka
Fully managed Service for running data streaming applications that leverage Apache Kafka

Control Plane
Provides control plane operations. Creates, updates, and deletes clusters as required

Data Plane
Use Kafka data plane operations for producing and consuming data streams

Existing Applications
Open-source versions of Apache Kafka allow support for existing applications, tools, and plug-ins



Amazon Managed Streaming for Apache Kafka

Concepts and Components

1 **Broker Nodes**
Specify the amount of broker nodes per AZ you want at time of cluster creation

2 **Zookeeper Nodes**
Zookeeper nodes are created for you

3 **Producers, Consumers, and topics**
Kafka data-plane operations allow creation of topics and ability to produce/consume data

4 **Flexible Cluster Operations**
Perform cluster operations with the console, AWS CLI, or APIs within any SDK

Resiliency in Amazon MSK

Automatics Recovery

Detection

Reduce Data

Time Required

After Recovery

Security and Logging

- Integration with Amazon KMS for SSE requirements
- Encryption at rest by default
- Encryption in transit between brokers in clusters
- Deliver broker logs to CloudWatch, S3, and Kinesis Data Firehose
- All API calls are logged to CloudTrail

Data Streaming vs Batch Processing (Activity)

Data Streaming	Batch Processing

VS

Checkpoint

You can use _____ to build a schema for your data, and _____ to query the data that's stored in S3.

1. Glue, Athena
2. EC2, SQS
3. EC2, Glue
4. Athena, Lambda

How much data can a Redshift database hold per cluster?

1. 1.6 PB
2. 1 TB
3. 16 PB
4. 10.6 PB

_____ provides real-time streaming of data.

1. Kinesis Data Firehose
2. Kinesis Data Streams
3. SQS
4. Kinesis Data Analytics

If you need to create a new streaming application requiring Apache Kafka as the primary component, which AWS service would be the best fit for this requirement?

1. Amazon Managed Streaming for Apache Kafka (MSK)
2. Amazon OpenStreaming Service
3. Amazon MQ
4. Amazon Kinesis

What is a valid use case for Amazon EMR?

1. Build real-time applications
2. Analyze data in Amazon S3 using standard SQL
3. Simplify data warehouse management
4. Extract, transform, and load (ETL) jobs

_____ allows you to transform data using SQL as it's being passed through Kinesis.

1. Redshift
2. Kinesis Data Analytics
3. DynamoDB
4. RDS

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